

YUVRAJ SINGH

ML Engineer — Computer Vision • Video Analytics • Activity Detection & Risk Automation

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Professional Summary

Computer Vision & ML Engineer with 2+ years of hands-on production experience in **video analytics, object detection, multi-task learning**, and **real-time inference systems**. Proven track record of independently designing and delivering end-to-end CV pipelines at **DRDO** (NVIDIA H100 multi-GPU, infrared video analytics) and **DeltaX AI** (industrial defect detection). Experienced in **YOLO (v5/v7/v8)**, **object tracking (DeepSORT / ByteTrack)**, **temporal video modeling**, **class-imbalance strategies**, and **production deployment** via FastAPI, Docker, and AWS. Strong research-application mindset backed by a **Springer publication** in infrared target detection. Seeking to own and scale a **workplace activity detection & risk assessment** platform from analysis through production.

Technical Skills

Core CV & Object Detection	YOLO (v5/v7/v8), Faster R-CNN, Detectron2, OpenCV, MPANet, YOLOv8-seg, Multi-Object Detection
Object Tracking	DeepSORT, ByteTrack, Multi-Object Tracking (MOT), IoU-based association, Kalman filter
Action & Activity Recognition	Temporal Classification, SlowFast, I3D, TimeSformer (applied knowledge), frame-level & sequence-level prediction, optical flow
Multi-Task & Video Analytics	Joint detection + classification, shared backbone architectures, temporal consistency, frame sampling strategies
Deep Learning Frameworks	PyTorch, TensorFlow, Keras, CNNs, Attention Mechanisms, Transfer Learning
Model Training & Optimization	Multi-GPU DDP (H100), Mixed Precision (AMP), OneCycleLR, Focal Loss, Weighted Loss, Mixup Augmentation, SMOTE
Class Imbalance Handling	Focal Loss, oversampling / undersampling, SMOTE, hard negative mining, threshold tuning
MLOps & Experiment Tracking	MLflow, Weights & Biases (W&B), Docker, Git, CI/CD pipelines, model versioning
Model Optimization & Deployment	ONNX, TensorRT, TorchScript, FP16 / quantization, FastAPI, TorchServe, batch inference
Cloud & Infrastructure	AWS (EC2, S3, SageMaker basics), Linux, CUDA, GPU inference optimization
Pose & Motion Estimation	MediaPipe Pose (keypoint detection), optical flow (Farneback, Lucas-Kanade), DeepFace
Data & Annotation	CVAT / Label Studio (annotation workflows), pandas, NumPy, SQL, Power BI

Professional Experience

ML / Computer Vision Engineer

DeltaX AI India — Industrial Vision Systems

April 2026 – Present
Hyderabad, India (Remote)

- Architecting a **multi-task learning pipeline** for simultaneous crack detection and severity classification on industrial metal-sheet video streams; shared EfficientNet-B3 backbone with independent detection and classification heads.
- Handling **severe class imbalance** (crack-positive frames <5% of dataset) via **focal loss**, hard-negative mining, and domain-aware data augmentation; targeting >80% recall at high precision.
- Optimising real-time video inference to **sub-50 ms latency** on GPU using **ONNX export**, **TorchScript**, and **FP16 mixed-precision** — pipeline processes 30+ FPS on production hardware.
- Deploying automated defect classification as a **FastAPI microservice** integrated into factory QA systems; containerised with Docker for reproducible environments.
- Tracking experiments, model checkpoints, and hyperparameter sweeps with **MLflow / W&B**; maintaining clean Git workflows for collaborative codebase.

AI/ML Engineer — Computer Vision (Video Analytics)

Defence Research and Development Organisation (DRDO)

Jun 2025 – Mar 2026
Dehradun, India

- **Independently designed** an end-to-end infrared video analytics pipeline for small target detection in low-SNR thermal sequences; selected MPANet + YOLO variants based on architecture trade-off analysis.
- Trained large-scale models on **NVIDIA H100 GPUs** using **PyTorch Distributed Data Parallel (DDP)**; managed multi-GPU job scheduling, checkpoint management, and failure recovery.
- Improved model performance from baseline through systematic root-cause analysis: recall **68% → 82%**, IoU **0.42 → 0.61** via architecture tuning, hard negative mining, and progressive training.
- Reduced inference latency by **25%** via CUDA kernel optimisation and **FP16 mixed-precision** inference without accuracy degradation — achieved **30+ FPS** on edge-deployable hardware.
- Engineered large-scale **temporal video preprocessing pipelines**: frame extraction, normalisation, augmentation, and annotation tooling for multi-thousand-hour thermal video datasets.
- Applied **temporal consistency constraints** across video segments; implemented frame-level and sequence-level prediction fusion for robust detection in cluttered scenes.
- Deployed real-time inference system; evaluated with domain-standard metrics (SCR, SNR, BS-F1) and confusion matrix analysis; documented architecture rationale and knowledge transfer materials.

Data & ML Engineer

Inevitable Infotech

Sep 2024 – May 2025

Lucknow, India

- Built SQL + Power BI dashboards for operational analytics; developed Python-based data processing pipelines for automated reporting workflows.

Key Projects

Workplace Occupant Monitoring System (ICMS) — Real-Time In-Cabin Analytics

2025–26

- Built an **In-Cabin Monitoring System** tested on Indian road conditions using NIR fisheye cameras; real-time occupant detection, pose estimation, and activity classification on a TI embedded board.
- Implemented **MediaPipe Pose** for human keypoint detection; combined with **object tracking** to ensure temporal consistency across frames on resource-constrained edge hardware.
- Addressed diverse occupant configurations (multiple objects, partial occlusion) using **multi-task learning**: joint detection + activity classification in a unified model.

Multi-Task Video Analytics Platform — Activity Detection, Tracking & Recognition

2024–25

- Engineered a unified **video analytics pipeline** combining **YOLOv8** object detection, **DeepSORT** multi-object tracking, **DeepFace**-based face recognition, and **optical flow** motion analysis for surveillance use cases — directly analogous to workplace activity monitoring.
- Achieved **94% recognition accuracy at 25 FPS** using **TorchScript** optimisation; designed temporal consistency logic across video segments to reduce false positives.
- Containerised full pipeline with **Docker**; exposed via REST API for downstream system integration.
- Implemented **frame-sampling strategies** and sequence-level aggregation for accurate **temporal classification** of multi-step actions across video clips.

Infrared Small Target Detection System — End-to-End Production Pipeline

2025–26

- Built low-SNR detection system (MPANet + YOLO variants) with **30+ FPS** real-time throughput; handled severe class imbalance in 50K+ bounding-box annotation dataset using weighted loss functions and hard negative mining.
- Achieved measurable metric gains: recall +14 pp, IoU +19 pp; published survey in **Springer** covering benchmarking methodology now adopted in ongoing DRDO work.

Industrial Crack Detection — Multi-Class, Class-Imbalanced Production System

2025–26

- Fine-tuned **EfficientNet-B3** for crack/no-crack classification; resolved domain gap between open-source training data and industrial steel images via targeted augmentation (**Mixup**, **OneCycleLR**).
- Achieved **99% validation accuracy**; deployed as **FastAPI REST endpoint** for real-time QA integration with Docker containerisation.

Credit Risk & Class Imbalance — Production ML API

2023–24

- Built classification model (Random Forest / Logistic Regression) achieving **87% ROC-AUC**; resolved extreme class skew via **SMOTE**, **focal loss weighting**, and **threshold tuning**.
- Deployed sub-100ms inference API with FastAPI + Docker; documented precision–recall trade-off analysis and failure mode investigation.

Research Publication

Infrared Small Target Detection — Survey & Evaluation

Springer (Published)

Surveyed traditional signal processing and deep learning methods for infrared target detection in video sequences. Evaluated

models using domain-standard metrics (SCR, SNR, BS-F1). Contributed novel benchmarking insights for low-SNR video detection — methodology adopted in ongoing DRDO work.

Education

B.Tech — Electronics & Communication Engineering

2020 – 2024

Dr. Ram Manohar Lohia Avadh University, Lucknow, India

Certifications

ML Deployment on AWS — Amazon Web Services • **AI Fundamentals** — Amazon Web Services • **SQL for Data Science** — Coursera

Key Achievements

- **Production impact at DRDO:** Independently raised infrared detection recall by 14 pp and IoU by 19 pp on a live defence system — documented and handed off with full architecture rationale.
- **Springer publication:** Contributed benchmarking methodology for low-SNR video detection now adopted in ongoing DRDO work.
- **End-to-end ownership:** Designed, trained, optimised, and deployed 6 + production CV systems (video analytics, detection, tracking, classification, RAG) across defence, industrial, and analytics domains — independently, with minimal guidance.
- **Multi-GPU scale:** Successfully managed large-scale model training on NVIDIA H100 clusters using PyTorch DDP; handled job scheduling, failure recovery, and checkpoint management.